

Topic: Interfacing of Wireless notice board.

Aim: Study of Arduino Nano and interfacing with external components for Wireless notice board with temperature sensor.

The concept of a Wireless Notice Board involved the use of embedded systems and wireless communication technologies to create a system capable of displaying messages or notices without the need for physical connections. We focused on developing a Wireless Notice Board system using Arduino and Bluetooth communication to transmit and receive messages remotely.

Connection of Components with Arduino nano

Arduino Vin	+ of breadboard
Arduino 5V	- of breadboard

LCD with I2C module

GND	- of breadboard
5V	+ of breadboard
SDA	A4 of Arduino
SCL	A5 of Arduino

Bluetooth module:

VCC	+ of breadboard
GND	- of breadboard
TXD	Arduino D2
RXD	Arduino D3

Temperature Sensor: DHT22

VCC (+ Pin)	Arduino 5V
GND (- Pin)	Arduino GND
Data Pin	Arduino D4

Code for connection:

```
#include <SoftwareSerial.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>
```

```
#define DHTPIN 4    // DHT11 module DATA pin connected to D4
#define DHTTYPE DHT11 // DHT11 sensor type
```

```
SoftwareSerial BT(2, 3); // RX, TX for Bluetooth
LiquidCrystal_I2C lcd(0x27, 16, 2); // LCD address 0x27 (change to 0x3F if needed)
DHT dht(DHTPIN, DHTTYPE); // Initialize DHT sensor
```

```

String lastMessage = "";

void setup() {
  Serial.begin(9600);
  BT.begin(9600);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("BT Notice Board");
  delay(3000);
  dht.begin(); // Start DHT11 sensor
  delay(2000); // Give sensor time to stabilize
}

void loop() {
  if (BT.available()) {
    lastMessage = ""; // Clear previous message
    while (BT.available()) {
      char c = BT.read();
      lastMessage += c;
      delay(50); // Ensures complete message is received
    }
    Serial.println("Received: " + lastMessage); // Debugging log
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(lastMessage.substring(0, 16));
  }

  // Read temperature from DHT11 module
  float temp = dht.readTemperature(false) / 2; // Read in Celsius

  // Debugging log
  Serial.print("Temperature: ");
  Serial.println(temp);

  lcd.setCursor(0, 1);
  if (isnan(temp)) {
    lcd.print("Temp: Error");
  } else {
    lcd.print("Temp: ");
    lcd.print(temp, 2);
    lcd.print(" C ");
  }

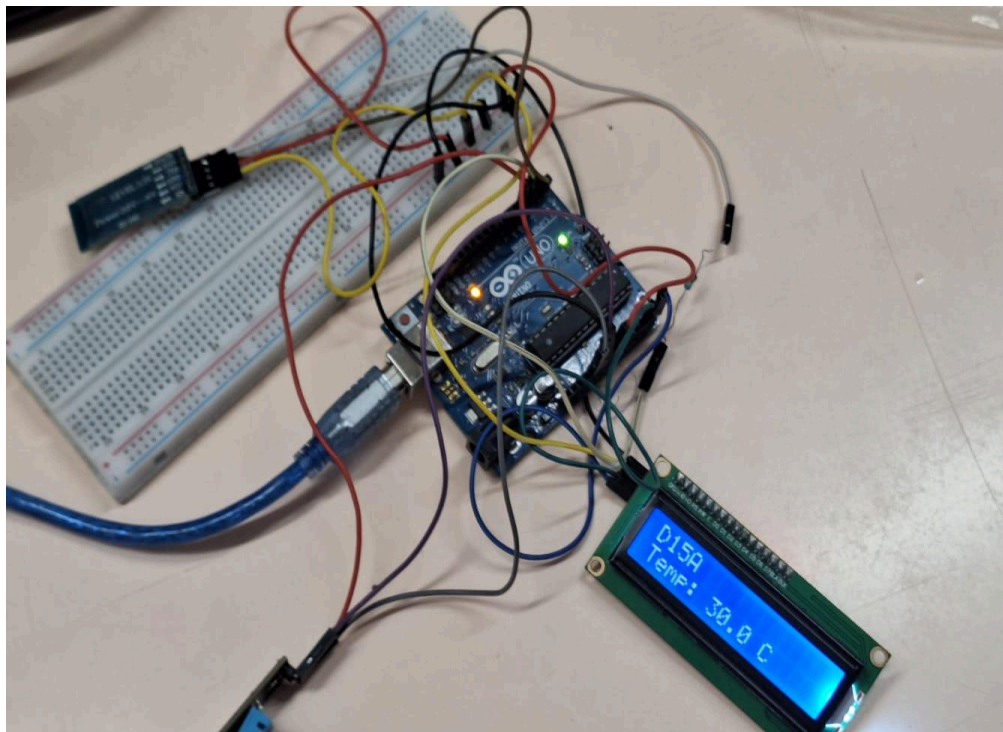
  delay(2500);
}

```

Libraries Used:

1. <SoftwareSerial.h>
This library allows serial communication on other digital pins of the Arduino, using software to replicate the functionality of hardware serial ports. It is commonly used to interface with serial devices such as Bluetooth modules when the hardware serial port is already in use.
2. <Wire.h>
The Wire library is used for I2C (Inter-Integrated Circuit) communication. It facilitates communication between the Arduino and I2C-compatible devices such as sensors, displays, and other microcontrollers.
3. <LiquidCrystal_I2C.h>
This library provides functions to interface with Liquid Crystal Displays (LCDs) that use the I2C communication protocol. It simplifies operations such as initializing the display, setting the cursor position, and printing text.
4. <DHT.h>
The DHT library is specifically designed to interface with DHT-series temperature and humidity sensors (e.g., DHT11, DHT22). It offers functions to initialize the sensor and to read temperature and humidity data.

Implementation:



**Conclusion:**

In this experiment, the Arduino Nano was successfully studied and utilized as the central microcontroller for interfacing with external components in a wireless notice board system. The system incorporated a Bluetooth module for wireless message transmission, a Liquid Crystal Display (LCD) for visual output, and a DHT11 sensor for real-time temperature monitoring. The Arduino Nano efficiently managed serial communication, sensor data acquisition, and dynamic display updates. This implementation demonstrates the capability of Arduino-based systems to integrate multiple modules for practical applications, such as real-time wireless communication and environmental monitoring, in a compact and cost-effective manner.